14.1.1 Adding nodes to the cluster

Each node in a cluster essentially runs an instance of Elasticsearch on a dedicated server. Nothing is stopping you from creating multiple nodes on a single server, but doing so defeats the purpose of data resiliency: if that server crashes, you'll lose all nodes on that server.

> **NOTE** It is advisable to deploy and run Elasticsearch on a dedicated server with as much computing power as your requirements demand, rather than bundling it with other applications—especially applications that are resource hungry.

When we launch an Elasticsearch server for the first time, a cluster is formed with a single node. This *single-node cluster* is the typical setup in development environments for testing and trying out products. Figure 14.1 shows a single-node cluster.

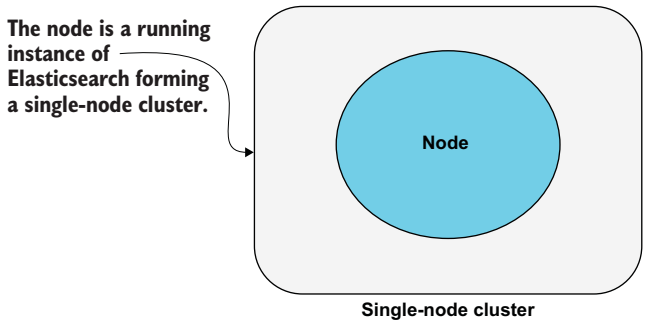


Figure 14.1 Single-node cluster

When we bring up more nodes (provided all nodes are christened with the same cluster name), they all join to make a *multi-node cluster*. Let's see how shards are created and distributed across the cluster as we keep adding nodes, from a single node to, say, three nodes.

Suppose we want to create an index called `chats` with one shard and one replica each. To do this, we need to define the number of shards and replicas during index creation by configuring the settings on the index.

Listing 14.1 Creating the `chats` index

```
PUT chats
{
  "settings": {
    "number_of_shards": 1,
    "number_of_replicas": 1
  }
}
```